Arunabha Majumder

+45 71 62 24 00 | Aalborg, Denmark
somrkmv1997@gmail.com | 97arunabhasit027@gmail.com
Portfolio: arunabha22.github.io/arunabha-majumder

November 2, 2026

NewOrbit Space, Reading, United Kingdom

Application: Satellite Mechanical Design Engineer

Dear NewOrbit Space team,

I am applying for the Satellite Mechanical Design Engineer role. I am a mechanical and mechatronics engineer doing my PhD at Aalborg University, with my thesis expected to be finished in March. Flying a satellite at 200 km is a bold idea, and what I like most is that it can only work if every bracket, panel and cable route is designed with care. That is the kind of detailed hardware work I enjoy.

For the last 3 years I have owned the mechanical design of a hybrid shoulder exoskeleton, from hand sketches to SolidWorks assemblies, drawings and working hardware. I optimised the design with a genetic algorithm, 3D printed and rebuilt the parts many times, routed and integrated the motor, encoder and force sensors, and measured the stiffness it provides on a physical test setup. Each round of testing fed back into the next design. It cut energy use by more than 25% and won the Best Paper Award at IFToMM ISRM 2026. I also designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms, estimated its stiffness with less than 10% error and quantified human arm impedance. In my Bachelor's I did the CAD, machining and assembly of a rehabilitation exoskeleton myself, so I know how a drawing turns into a real part.

SolidWorks is my main tool (CSWA), and I am used to keeping assemblies tidy while a design keeps changing. I work closely with control, electronics and software colleagues, and I explain my design choices clearly, often with a quick sketch, since I also hold a Diploma in Fine Arts.

I want to be honest about the gaps. I have not designed space hardware, and I have not yet used GD&T or PLM systems like 3DEXPERIENCE in a formal change process. I learn quickly, and I would really like to learn these from an experienced team building real satellites.

Thank you for considering my application. You can see my work at arunabha22.github.io/arunabha-majumder.

Kind regards,
Arunabha Majumder